

onSaveInstanceState Vs SavedStateHandle In Android Development





Overview

Android provides two mechanisms for saving and restoring UI-related data during configuration changes or process death:

- onSaveInstanceState: A lifecycle method used in Activities or Fragments
- SavedStateHandle: A Jetpack tool used within ViewModels

Both aim to retain UI-related data, but they serve different use cases and lifecycles.



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Feature	onSaveInstanceState()	SavedStateHandle
Scope	Activity / Fragment	ViewModel
Purpose	Save temporary UI state	Persist ViewModel-related state
Called When	Before activity is destroyed (e.g., rotation)	Automatically restores state during ViewModel recreation
Persistence	Short-lived (for configuration changes)	Longer-lived (supports process death recovery)

Example:

onSaveInstanceState()

```
override fun onSaveInstanceState(outState: Bundle) {  
    outState.putString("text", editText.text.toString())  
}
```

SavedStateHandle in ViewModel

```
class MyViewModel(savedStateHandle: SavedStateHandle) : ViewModel() {  
    val selectedItem = savedStateHandle.getLiveData<String>("selectedItem")  
}
```

